

Enhancing Pre-trained Diffusion Models with Reinforcement Learning and Adversarial Reward Functions

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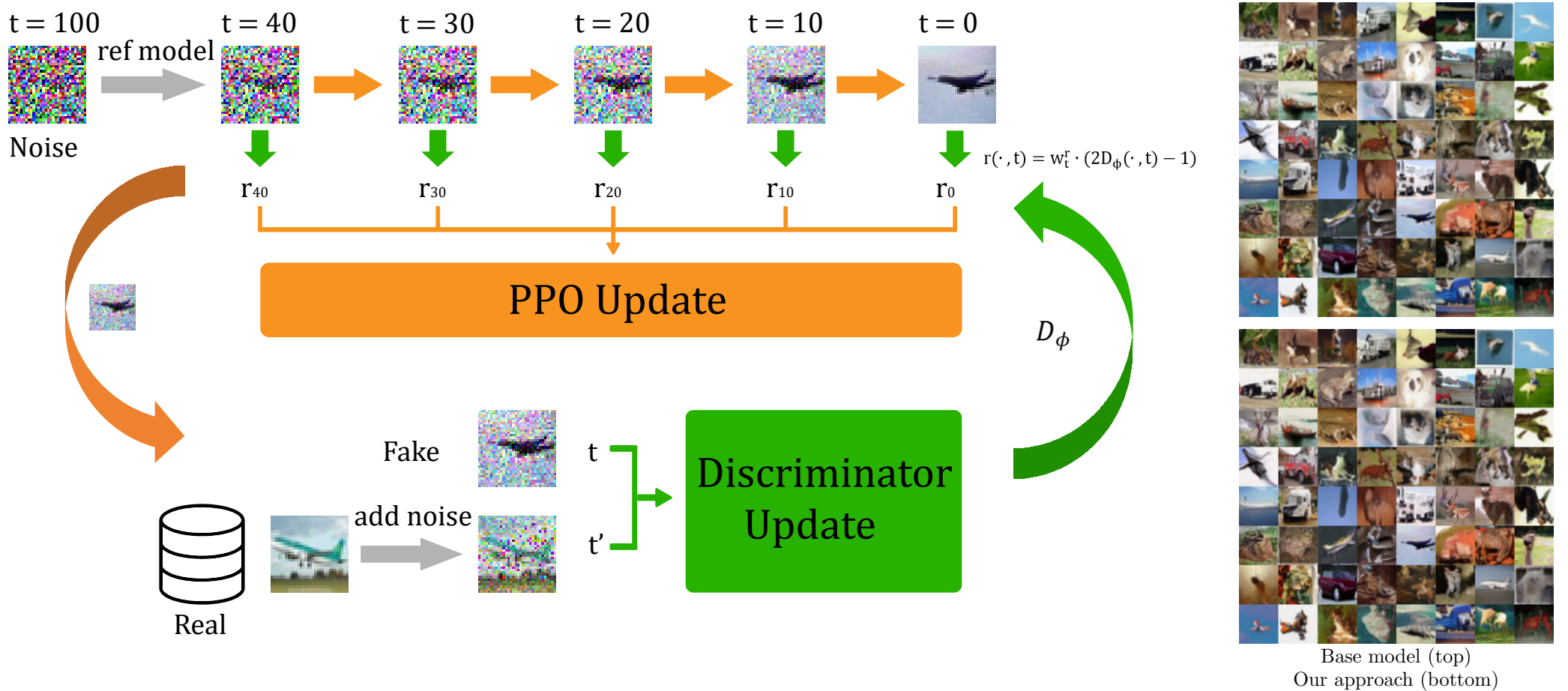


Figure 1: Left: Overview of our framework. We formulate the reverse diffusion process as a Markov Decision Process (MDP) and use an adversarial discriminator to provide per-timestep rewards. The diffusion model is trained using Proximal Policy Optimization (PPO). Right: Generated samples comparison. Top: Base model. Bottom: Our approach.

Why RL for Diffusion Models?

- Diffusion models are strong generators, but fixed objectives may limit sample quality.
- Prior improvements: distillation, supervised loss—but still indirect for perceptual quality.
- **Our solution:** Directly optimize perceptual quality by treating diffusion as a Markov Decision Process (MDP) and training with RL and adversarial rewards!
- **Plug-and-play:** No architecture/inference change, works with existing pipelines.

How does it work?

We propose using reinforcement learning to fine-tune diffusion models with adversarial rewards. Our approach formulates the reverse diffusion process as a Markov Decision Process (MDP) and uses a discriminator to evaluate image quality during generation.

MDP Formulation: We formalize the diffusion process $x_{T'} \rightarrow x_0$ as an MDP where:

- **States** s_t : Noisy images $(\mathbf{x}_t, t)$ at timestep t
- **Actions** a_t : Predicted noise $\epsilon_\theta(\mathbf{x}_t, t)$
- **Rewards** r_t : Based on discriminator output: $r_t = w_t^r \cdot (2D(\mathbf{x}_t, t) - 1)$

Discriminator: Uses a weighted GAN loss with gradient penalties:

$$\mathcal{L}_D(\phi) = \mathbb{E}_t \left[w_t^d \cdot \mathbb{E}_{\substack{x_{T'} \sim p_{T'} \\ x_t \sim \text{Denoise}(x_{T'}, \epsilon_\theta, t)}} [\log D_\phi(x_t, t)] \right] + \mathbb{E}_t \left[w_t^d \cdot \mathbb{E}_{\substack{x_0 \sim \mathcal{D}_{\text{real}} \\ x_t \sim \text{AddNoise}(x_0, t)}} [\log (1 - D_\phi(x_t, t))] \right].$$

We regularize the discriminator using R_1 and R_2 gradient penalties:

$$R_1(\phi) = \frac{\gamma}{2} \mathbb{E}_{\text{real}} [\|\nabla_x D_\phi(x_t, t)\|^2], \quad R_2(\phi) = \frac{\gamma}{2} \mathbb{E}_{\text{gen}} [\|\nabla_x D_\phi(x_t, t)\|^2]$$

where R_1 is applied to real samples and R_2 to generated ones. These penalties stabilize training by enforcing smoothness on the discriminator.

Optimization: We employ Proximal Policy Optimization (PPO) with:

- DDIM sampling for deterministic transitions
- KL divergence tracking to prevent model degradation
- Advantage normalization for stable training
- Since PPO needs a nondeterministic action distribution, we perturb the predicted noise (action) at a small fixed scale σ while preserving the norm of predicted noise:

$$\epsilon_{\text{perturb}} = \sqrt{1 - \sigma^2} \epsilon + \sigma \epsilon_{\text{noise}}, \quad \epsilon_{\text{noise}} \sim \mathcal{N}(0, I).$$

Stabilization Techniques:

- Discriminator freezing when it becomes too powerful
- Reset mechanism to restore best checkpoint if FID deteriorates
- Exponential Moving Average (EMA) model $\theta' \leftarrow \beta\theta' + (1 - \beta)\theta$ for robust sampling

Training Loop:

1. Generate trajectories by denoising $x_{T'}$
2. Compute discriminator-based rewards
3. Update discriminator with gradient penalties
4. Update policy using PPO with KL constraint
5. Reset if FID score deteriorates

Results

Our base model reaches an FID score of 18.26 for $T = 100$ and 15.75 for $T = 1000$. Our approach achieves an FID score of **13.11** for $T = 100$ and 16.81 for $T = 1000$. Notably, our approach with $T = 100$ has a lower FID score than the base model with $T = 1000$, indicating that our approach is able to improve both sample efficiency and sample quality.

Takeaway: Our RL+adversarial approach improves both sample efficiency and quality!

Ablation: Reset prevents collapse, R_1/R_2 and EMA reduce FID variance and improve stability.

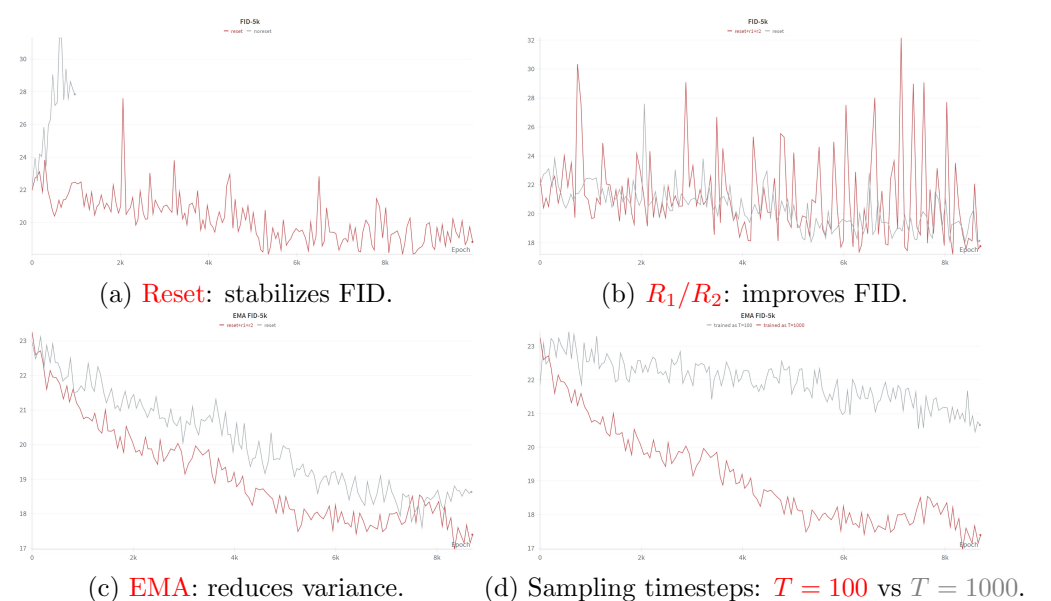


Figure 2: Ablation: Each component boosts stability or quality.